

AERO70015/96012 Mathematics 3 Problem Set #5

Problem Sheet 5

5.1 PR5_1

Sketch the image of the quarter circle $|z| < 1$, $0 < \arg(z) < \pi/2$ under each of the following transformations:

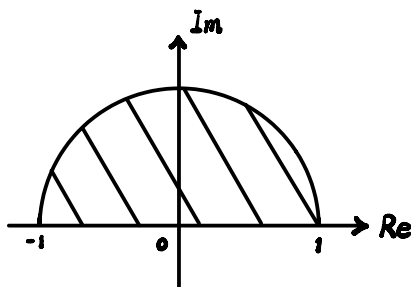
- (a) $w = z^2$
- (b) $w = iz$
- (c) $w = 1/z$
- (d) $w = \ln z$

$$(a) \quad w = z^2 = |z|^2 e^{i \arg(z)} = |z|^2 e^{i[2 \arg(z)]}$$

$$\text{where } |z|^2 < 1$$

$$0 < 2 \arg(z) < \pi$$

$\therefore$

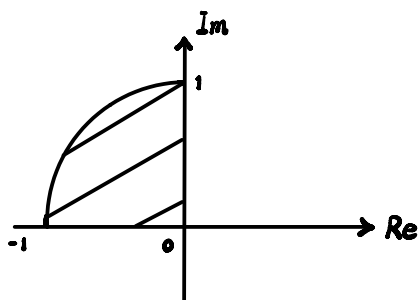


$$(b) \quad w = iz = i|z| e^{i \arg(z)} = |z| e^{i \arg(z)} e^{i \frac{\pi}{2}} = |z| e^{i[\arg(z) + \frac{\pi}{2}]}$$

$$\text{where } |z| < 1$$

$$\frac{\pi}{2} < \arg(z) + \frac{\pi}{2} < \pi$$

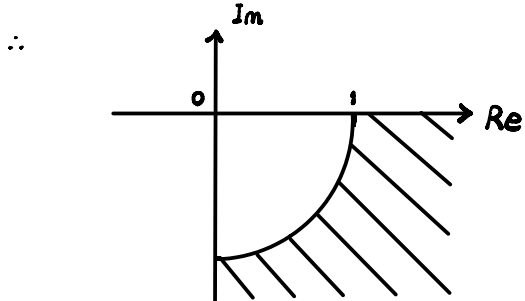
$\therefore$



$$(c) \quad w = \frac{1}{z} = \frac{1}{|z|} e^{-i \arg(z)}$$

$$\text{where } \frac{1}{|z|} > 1$$

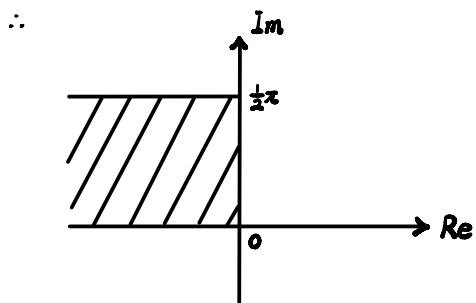
$$-\frac{1}{2}\pi < -\arg(z) < 0$$



$$(d) \quad w = \ln z = \ln |z| + i \arg(z)$$

$$\text{where } \ln |z| < 0$$

$$0 < \arg(z) < \frac{1}{2}\pi$$



5.2 PR5_2

Sketch the region

$$|x - y| < \pi/4, \quad x > |y|$$

in the (x, y) -plane, and its image under the transformation $w = u + i v = (x + i y)^2 = z^2$.

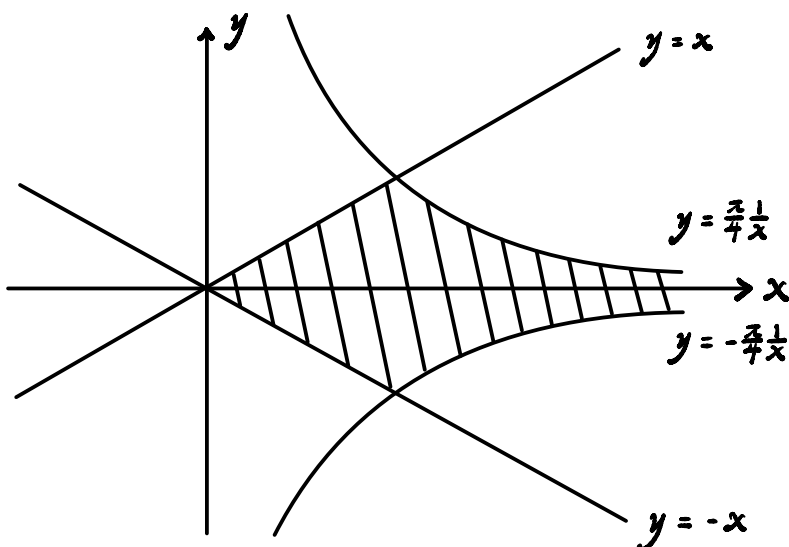
The function $\phi(x, y)$ is bounded and harmonic throughout this region. Find ϕ in terms of x and y in each of the following cases:

- (a) $\phi = 1$ on $x - y = \pi/4, x > y$; $\phi = 0$ on $x - y = -\pi/4, x < -y$; gradient of ϕ normal to $x = \pm y$ is zero on $x = \pm y$
- (b) $\phi = 0$ on $x - y = \pm\pi/4, x > |y|$; $\phi = \cos(2 x - y)$ on $x = \pm y, |x - y| < \pi/4$

$$-\frac{\pi}{4} < x - y < \frac{\pi}{4} \quad \therefore -\frac{\pi}{4} \frac{1}{x} < y < \frac{\pi}{4} \frac{1}{x}$$

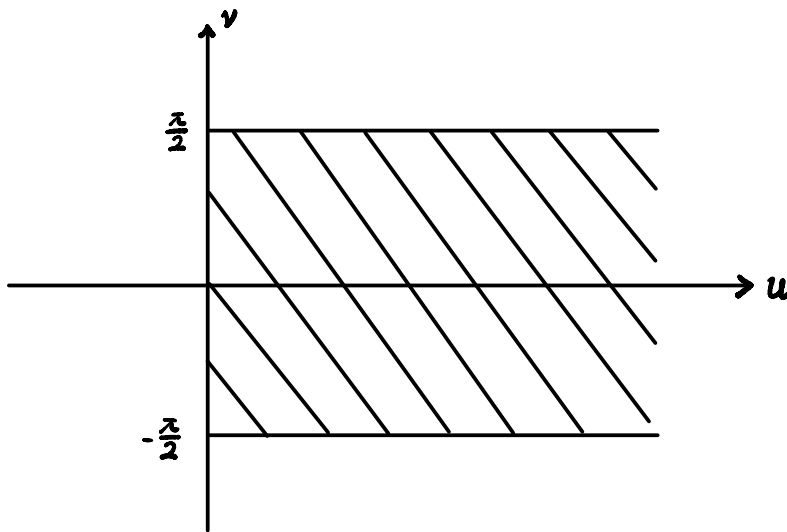
$$y < x < -y \quad \therefore y < x \text{ and } y > -x$$

Therefore, the bounded region will be like



After transformation,

$$w = x^2 - y^2 + 2ixy = u + iv$$



$$(a) \quad \phi = 1 \text{ on } u > 0, v = \frac{\pi}{2}$$

$$\phi = 0 \text{ on } u > 0, v = -\frac{\pi}{2}$$

$$\frac{d\phi}{du} = 0 \text{ on } u = 0$$

Also, ϕ independent of u along $v = \pm \frac{\pi}{2}$

$$\therefore \nabla^2 \phi = \frac{\partial^2 \phi}{\partial u^2} + \frac{\partial^2 \phi}{\partial v^2} = \frac{\partial^2 \phi}{\partial v^2} = 0$$

$$\therefore \phi(u, v) = \phi(v) = Av + B$$

$$\therefore \begin{cases} 1 = \frac{\pi}{2}A + B \\ 0 = -\frac{\pi}{2}A + B \end{cases}$$

$$\therefore A = \frac{1}{\pi}, B = \frac{1}{2}$$

$$\therefore \phi(v) = \frac{1}{\pi}v + \frac{1}{2}$$

$$\therefore \phi(x, y) = \frac{2xy}{\pi} + \frac{1}{2}$$

$$b) \phi = 0 \text{ on } u > 0, v = \pm \frac{\pi}{2}$$

$$\phi = \cos(2xy) = \cos v \text{ on } u = 0, -\frac{\pi}{2} < v < \frac{\pi}{2}$$

$$\therefore \text{Let } \phi(u, v) = F(u)G(v) \quad \therefore \nabla^2 \phi = F''(u)G(v) + F(u)G''(v) = 0$$

$$\text{Let } \frac{F''(u)}{F(u)} = -\frac{G''(v)}{G(v)} = k^2$$

$$\text{Let } F(u) = Ae^{ku} + Be^{-ku}, G(v) = C\cos(kv) + D\sin(kv)$$

$$\therefore \phi = 0 \text{ on } u > 0, v = \pm \frac{\pi}{2} \quad \therefore D = 0$$

$$\therefore \text{region is bounded as } u \rightarrow \infty \quad \therefore A = 0$$

$$\text{Let } C = 1 \quad \therefore \phi(u, v) = \sum_{k=1}^{\infty} \cos(kv) (B_k e^{-ku}) \text{ for } k = 0, 1, 2, \dots$$

$$\therefore \phi = \cos v \text{ on } u = 0, -\frac{\pi}{2} < v < \frac{\pi}{2} \quad \therefore B_1 = 1, B_k|_{k \neq 1} = 0$$

$$\therefore \phi(u, v) = e^{-u} \cos v \quad \therefore \phi(x, y) = e^{-(x^2+y^2)} \cos(2xy)$$

5.3 PR5_3

Show that the function

$$w = \frac{z-i}{z+i}$$

maps the upper half-plane $y > 0$ onto the interior of the circle $|w| < 1$.

Solve

$$\phi_{xx} + \phi_{yy} = 0 \quad (y > 0)$$

subject to the boundary conditions

$$\phi(x, 0) = (x^2 - 1)/(x^2 + 1); \quad \phi \text{ bounded at infinity,}$$

by transforming to the (u, v) -plane, where $w = u + i v$.

$$w = \frac{z-i}{z+i}$$

$$\therefore (z+i)w = z-i$$

$$\therefore z(w-1) = -i(w+1)$$

$$\therefore z = \frac{-i(w+1)}{w-1}$$

$$\begin{aligned}\therefore x+iy &= \frac{-i(u+1+iv)}{u-1+iv} = \frac{v-i(u+1)}{(u-1)+iv} = \frac{[v-i(u+1)][(u-1)-iv]}{(u-1)^2+v^2} \\ &= \frac{(u-1)v - (u+1)v - i(u^2-1+v^2)}{(u-1)^2+v^2} \\ &= -\frac{2v}{(u-1)^2+v^2} + i \frac{1-(u^2+v^2)}{(u-1)^2+v^2}\end{aligned}$$

$$\therefore y = \frac{1-(u^2+v^2)}{(u-1)^2+v^2} > 0 \quad \text{when } u^2+v^2 = |w|^2 < 1$$

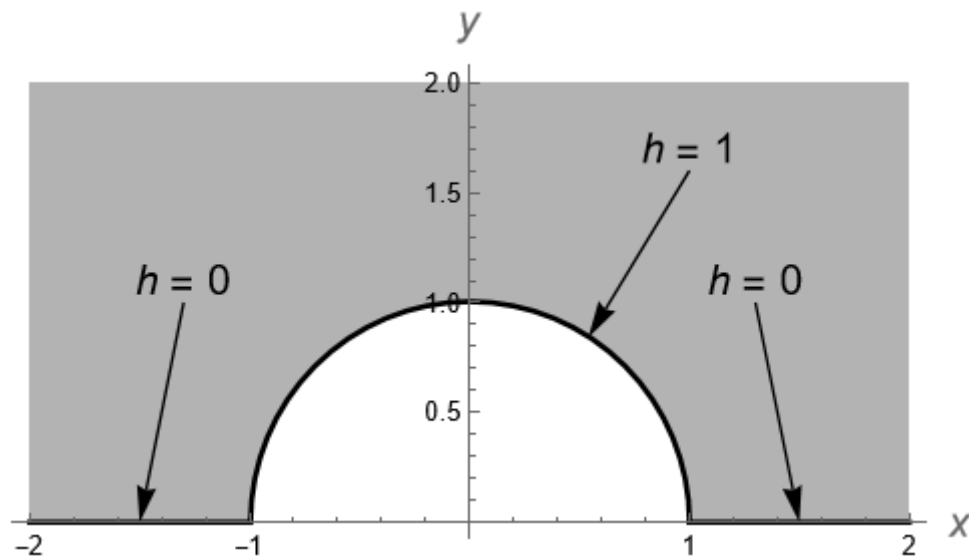
$$\begin{aligned}\therefore u+iv &= \frac{x+i(y-1)}{x+i(y+1)} = \frac{x^2+y^2-1+ix(y-1)-ix(y+1)}{x^2+(y+1)^2} \\ &= \frac{x^2+y^2-1}{x^2+(y+1)^2} - i \frac{2x}{x^2+(y+1)^2}\end{aligned}$$

$$\therefore w(x,0) = \frac{x^2-1}{x^2+1} - i \frac{2x}{x^2+1}$$

$$\therefore \phi(x,0) = \frac{x^2-1}{x^2+1} = u \quad \text{on } |w|=1 \quad \text{which satisfied } \nabla^2 \phi = 0$$

$$\therefore \phi(x,y) = \frac{x^2+y^2-1}{x^2+(y+1)^2}$$

5.4 PR5_4



The variable h satisfies Laplace's equation,

$$\frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial y^2} = 0,$$

on the shaded region shown in the figure, with boundary conditions as in the diagram.

- Determine the region D' in the w -plane to which the shaded region is transformed under the conformal map $w = \frac{z-1}{z+1}$, and specify the boundary conditions on this transformed region.
- Solve Laplace's equation for h on D' .
- Write u and v as functions of x and y .
- Hence, obtain the solution on the original shaded region.

$$(a) \quad w = \frac{z-1}{z+1}$$

$$\therefore u + iv = \frac{(x-1) + iy}{(x+1) + iy} = \frac{(x^2-1) + y^2 + 2iy}{(x+1)^2 + y^2} = \frac{(x^2+y^2)-1}{(x+1)^2 + y^2} + i \frac{2y}{(x+1)^2 + y^2}$$

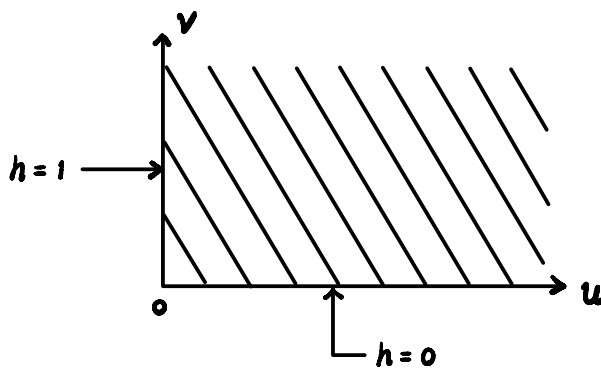
$$\text{At } h=1, (x^2+y^2) > 1 \quad \therefore u > 0$$

$$\therefore w = \frac{z-1}{z+1} \quad \therefore z(w-1) = -(w+1) \quad \therefore z = -\frac{w+1}{w-1}$$

$$\therefore x + iy = -\frac{(u+1) + iv}{(u-1) + iv} = -\frac{u^2-1+v^2-2iv}{(u-1)^2 + v^2} = \frac{1-(u^2+v^2)}{(u-1)^2 + v^2} + i \frac{2v}{(u-1)^2 + v^2}$$

$$\text{At } h=0, y > 0 \quad \therefore v > 0$$

As shown in diagram,



(b) The boundary condition depend on $\theta = \arg w = \tan^{-1}(\frac{v}{u})$

$$\therefore \text{let } \theta = \tan^{-1}(\frac{v}{u}) + C$$

$$\frac{\partial h}{\partial u} = \frac{dh}{d\theta} \frac{d\theta}{du} = \frac{1}{1+(\frac{v}{u})^2} (-\frac{v}{u^2}) \frac{dh}{d\theta} = -\frac{v}{u^2+v^2} \frac{dh}{d\theta}$$

$$\frac{\partial h}{\partial v} = \frac{dh}{d\theta} \frac{d\theta}{dv} = \frac{1}{1+(\frac{v}{u})^2} (\frac{1}{u}) \frac{dh}{d\theta} = \frac{u}{u^2+v^2} \frac{dh}{d\theta}$$

$$\begin{aligned} \therefore \frac{\partial^2 h}{\partial u^2} &= \frac{\partial}{\partial u} \left(-\frac{v}{u^2+v^2} \frac{dh}{d\theta} \right) = \frac{2uv}{u^2+v^2} \frac{dh}{d\theta} - \frac{v}{u^2+v^2} \frac{d}{du} \left(\frac{dh}{d\theta} \right) \\ &= \frac{2uv}{u^2+v^2} \frac{dh}{d\theta} + \frac{v^2}{(u^2+v^2)^2} \frac{d^2 h}{d\theta^2} \end{aligned}$$

$$\begin{aligned} \frac{\partial^2 h}{\partial v^2} &= \frac{\partial}{\partial v} \left(\frac{u}{u^2+v^2} \frac{dh}{d\theta} \right) = -\frac{2uv}{u^2+v^2} \frac{dh}{d\theta} + \frac{u}{u^2+v^2} \frac{d}{dv} \left(\frac{dh}{d\theta} \right) \\ &= -\frac{2uv}{u^2+v^2} \frac{dh}{d\theta} + \frac{u^2}{(u^2+v^2)^2} \frac{d^2 h}{d\theta^2} \end{aligned}$$

$$\therefore \nabla^2 h = \frac{\partial^2 h}{\partial u^2} + \frac{\partial^2 h}{\partial v^2} = \frac{1}{u^2+v^2} \frac{d^2 h}{d\theta^2} = 0$$

$$\therefore \frac{d^2 h}{d\theta^2} = 0$$

$$\therefore h = A\theta + B = A \tan^{-1}(\frac{v}{u}) + B$$

$$\text{At } v=0, h=B=0$$

$$\text{At } u=0, h = \frac{\pi}{2} A = 1 \quad \therefore A = \frac{2}{\pi}$$

$$\therefore h = \frac{2}{\pi} \tan^{-1} \left(\frac{v}{u} \right)$$

(c) As shown in (a),

$$u = \frac{(x^2 + y^2) - 1}{(x+1)^2 + y^2}$$

$$v = \frac{2y}{(x+1)^2 + y^2}$$

$$(d) h = \frac{2}{\pi} \tan^{-1} \left(\frac{2y}{x^2 + y^2 - 1} \right)$$

5.5 PR5_5

(a) Show that the Joukowski transformation,

$$w = z + \frac{c^2}{z},$$

where c is a positive real number, transforms the region outside and on the circle $x^2 + y^2 = a^2$ (where $a > c$) onto the region outside and on the ellipse $u^2/A^2 + v^2/B^2 = 1$, where $w = u + i v$, $z = x + i y$. Write down A and B in terms of x and y .

(b) It can be shown that if a steady flow has complex potential $\Omega(z)$ about a fixed body with boundary $\mathcal{C}$, then the moment M of the forces exerted on the body by the fluid is given by

$$M = -\frac{\rho}{2} \operatorname{Re} \left(\oint_{\mathcal{C}} w \left[\frac{d\Omega}{dw} \right]^2 dw \right).$$

Given that for flow at an angle α past a circular cylinder of radius a , the complex potential is of the form

$$\Omega(z) = z e^{-i\alpha} + \frac{a^2}{z} e^{i\alpha},$$

find the moment of the forces resulting from flow at an angle α past an *elliptical* cylinder with boundary $u^2/A^2 + v^2/B^2 = 1$ in the (u, v) -plane. (Hint: use the Joukowski transformation to transform the expression for M into an integral around a circle in the complex z -plane, then use the Residue Theorem.)

$$(a) \quad w = re^{i\theta} + \frac{c^2}{r} e^{-i\theta}$$

$$= \left(r + \frac{c^2}{r}\right) \cos\theta + i \left(r - \frac{c^2}{r}\right) \sin\theta = u + iv$$

$$\therefore |w|^2 = w\bar{w} = \left(r + \frac{c^2}{r}\right)^2 \cos^2\theta + \left(r - \frac{c^2}{r}\right)^2 \sin^2\theta$$

$$\therefore \frac{\left(r + \frac{c^2}{r}\right)^2 \cos^2\theta}{\left(r + \frac{c^2}{r}\right)^2} + \frac{\left(r - \frac{c^2}{r}\right)^2 \sin^2\theta}{\left(r - \frac{c^2}{r}\right)^2} = \frac{u^2}{A^2} + \frac{v^2}{B^2} = 1$$

$$\therefore A = r + \frac{c^2}{r} = \sqrt{x^2 + y^2} + \frac{c^2}{\sqrt{x^2 + y^2}} = a + \frac{c^2}{a}$$

$$B = r - \frac{c^2}{r} = \sqrt{x^2 + y^2} - \frac{c^2}{\sqrt{x^2 + y^2}} = a - \frac{c^2}{a}$$

$$(b) \quad \frac{dw}{dz} = 1 - \frac{c^2}{z^2}, \quad \frac{d\theta}{dz} = e^{-i\alpha} - \frac{a^2}{z^2} e^{i\alpha}$$

$$\therefore \oint_C w \left(\frac{d\theta}{dw}\right)^2 dw = \oint_C w \left(\frac{d\theta}{dz} \frac{dz}{dw}\right)^2 \frac{dw}{dz} dz$$

$$= \oint_C w \left(\frac{d\theta}{dz}\right)^2 \left(\frac{dw}{dz}\right)^{-1} dz$$

$$= \oint_C \left(\frac{z^2 + c^2}{z}\right) \left(e^{-i\alpha} - \frac{a^2}{z^2} e^{i\alpha}\right)^2 \left(\frac{z^2}{z^2 - c^2}\right) dz$$

$$= \oint_C \frac{z(z^2 + c^2)}{z^2 - c^2} \left(e^{-2i\alpha} - 2\frac{a^2}{z^2} + \frac{a^4}{z^4} e^{2i\alpha}\right) dz$$

$$\text{let } f(z) = \frac{z(z^2 + c^2)}{z^2 - c^2} \left(e^{-2i\alpha} - 2\frac{a^2}{z^2} + \frac{a^4}{z^4} e^{2i\alpha}\right)$$

$$\therefore \text{Res}_{z=c} = \lim_{z \rightarrow c} \frac{z(z^2 + c^2)}{z - c} \left(e^{-2i\alpha} - 2\frac{a^2}{z^2} + \frac{a^4}{z^4} e^{2i\alpha}\right)$$

$$= c^2 \left(e^{-2i\alpha} - 2\frac{a^2}{c^2} + \frac{a^4}{c^4} e^{2i\alpha}\right)$$

$$\therefore \text{Res}_{z=-c} = \lim_{z \rightarrow -c} \frac{z(z^2 + c^2)}{z - c} \left(e^{-2i\alpha} - 2\frac{a^2}{z^2} + \frac{a^4}{z^4} e^{2i\alpha}\right)$$

$$= c^2 (e^{-2i\alpha} - 2 \frac{a^2}{c^2} + \frac{a^*}{c^*} e^{2i\alpha})$$

$$\text{At } z_0 = 0,$$

$$f(z) = \frac{-z(z^2 + c^2)}{c^2} \frac{1}{1 - \frac{z^2}{c^2}} (e^{-2i\alpha} - 2 \frac{a^2}{z^2} + \frac{a^*}{z^*} e^{2i\alpha})$$

$$= -(\frac{z^3}{c^2} + z) (1 + \frac{z^2}{c^2} + \frac{z^4}{c^4} + \dots) (e^{-2i\alpha} - 2 \frac{a^2}{z^2} + \frac{a^*}{z^*} e^{2i\alpha})$$

$$\therefore \text{Res}_{z=0} = C_{-1} = 2a^2 - 2 \frac{a^*}{c^2} e^{2i\alpha}$$

$$\therefore \frac{1}{2\pi i} \oint_c w \left(\frac{dw}{dz} \right)^2 dz = \text{Res}_{z=c} + \text{Res}_{z=-c} + \text{Res}_{z=0}$$

$$= 2c^2 (e^{-2i\alpha} - 2 \frac{a^2}{c^2} + \frac{a^*}{c^*} e^{2i\alpha}) + 2a^2 - 2 \frac{a^*}{c^2} e^{2i\alpha}$$

$$= 2c^2 e^{-2i\alpha} - 2a^2$$

$$= -2a^2 + 2c^2 \cos(2\alpha) - 2ic^2 \sin(2\alpha)$$

$$\therefore \text{Re} \left\{ \oint_c w \left(\frac{dw}{dz} \right)^2 dz \right\} = 4\pi c^2 \sin(2\alpha)$$

$$\therefore M = -\frac{1}{2} \rho \text{Re} \left\{ \oint_c w \left(\frac{dw}{dz} \right)^2 dz \right\}$$

$$= -2c^2 \rho \pi \sin(2\alpha)$$

$$\therefore c^2 = \frac{1}{4} (A^2 - B^2)$$

$$\therefore M = -\frac{1}{2} (A^2 - B^2) \rho \pi \sin(2\alpha)$$

5.6 2017 Exam

(a) The complex function

$$w = \frac{z-i}{z}$$

maps a point $z = x + iy$ in the complex (x, y) -plane to a point $w = u + iv$ in the complex (u, v) -plane. Find z in terms of w , and hence express x and y in terms of u and v .

Find the curves in the (u, v) -plane to which the following curves in the (x, y) -plane are mapped:

- (i) C_1 , the imaginary axis;
- (ii) C_2 , the circle centre $\left(0, \frac{1}{2}\right)$, radius $\frac{1}{2}$;
- (iii) C_3 , the circle centre $\left(-\frac{1}{2}, 0\right)$, radius $\frac{1}{2}$.

Sketch the finite region D in the (x, y) -plane that is bounded by all three of the curves C_1 , C_2 and C_3 . Sketch too the region D' in the (u, v) -plane to which D is mapped.

(b) The real function $\phi(x, y)$ is defined over the region D defined in (a), and satisfies Laplace's equation,

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$$

in the interior of D . It is also bounded, and satisfies the following boundary conditions:

- along C_1 , $\phi(x, y) = 0$;
- along C_3 , $\phi(x, y) = 0$;
- along C_2 , $\phi(x, y) = -\sin\left(\frac{\pi x}{x^2 + y^2}\right)$.

You may assume that ϕ also satisfies Laplace's equation, with respect to u and v , in the interior of D' as defined in (i).

State the boundary conditions in the (u, v) -plane, and hence, by separation of variables, obtain ϕ as a function of u and v . Hence obtain ϕ as a function of x and y .

$$(a) \quad u + iv = \frac{x + i(y-1)}{x + iy} = \frac{x^2 + y(y-1) - ix}{x^2 + y^2} = \left(1 - \frac{y}{x^2 + y^2}\right) + i\left(-\frac{x}{x^2 + y^2}\right)$$

$$w = \frac{z-i}{z} \quad \therefore zw = z - i \quad \therefore z(w-1) = -i \quad \therefore z = -\frac{i}{w-1}$$

$$\therefore x + iy = -\frac{i}{(u-1) + iv} = -\frac{i[(u-1) - iv]}{(u-1)^2 + v^2} = -\frac{v}{(u-1)^2 + v^2} + i\left[\frac{-(u-1)}{(u-1)^2 + v^2}\right]$$

$$(i) \quad C_1 : v = 0$$

$$(ii) \quad C_2 \text{ for } x^2 + \left(y - \frac{1}{2}\right)^2 = \frac{1}{4} \quad \therefore x^2 + y^2 = y$$

$$\therefore \frac{v^2}{[(u-1)^2 + v^2]} + \frac{(u-1)^2}{[(u-1)^2 + v^2]} = \frac{-(u-1)}{(u-1)^2 + v^2}$$

$$\therefore \frac{1}{(u-1)^2 + v^2} + \frac{(u-1)}{(u-1)^2 + v^2} = 0$$

$$\therefore \frac{u}{(u-1)^2 + v^2} = 0$$

$$\therefore C_2 : u = 0$$

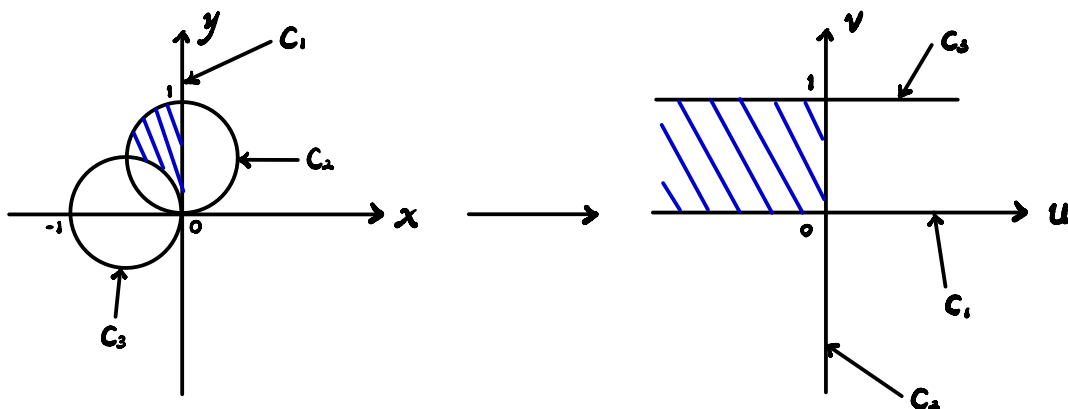
ii) C_3 for $(x^2 + \frac{1}{2})^2 + y^2 = \frac{1}{4} \quad \therefore x^2 + y^2 + x = 0$

$$\therefore \frac{v^2}{[(u-1)^2 + v^2]} + \frac{(u-1)^2}{[(u-1)^2 + v^2]^2} - \frac{v}{(u-1)^2 + v^2} = 0$$

$$\therefore \frac{1}{(u-1)^2 + v^2} - \frac{v}{(u-1)^2 + v^2} = 0$$

$$\therefore \frac{1-v}{(u-1)^2 + v^2} = 0$$

$$\therefore C_3 : v = 1$$



(b) Along C_1 , $v = 0$, $\phi = 0$

Along C_3 , $v = 1$, $\phi = 0$

Along C_2 , $u = 0$, $\phi = -\sin\left(\frac{\pi x}{x^2 + y^2}\right) = \sin\left(-\frac{\pi x}{x^2 + y^2}\right)$

$$\therefore u = 1 - \frac{y}{x^2 + y^2}$$

$$v = -\frac{x}{x^2 + y^2}$$

$$\therefore \text{Along } C_2, u=0, \phi = \sin(\pi v)$$

$$\text{Let } \phi(u, v) = F(u) G(v)$$

$$\therefore \nabla^2 \phi = 0$$

$$\therefore \frac{F''}{F} = -\frac{G''}{G} = -k^2$$

$$\text{Let } F(u) = Ae^{ku} + Be^{-ku}, G(v) = C \cos(kv) + D \sin(kv)$$

$$\text{Along } C_1, v=0, \phi=0 \quad \therefore C=0$$

$$\text{Along } C_3, v=1, \phi=0 \quad \therefore D \sin(k) = 0 \quad \therefore k = n\pi \text{ where } n = 0, 1, 2, \dots$$

$$\therefore \text{region bounded till } u \rightarrow \infty \quad \therefore A=0$$

$$\text{Let } D=1, \text{ therefore } \phi(u, v) = \sum_{n=0}^{\infty} B_n e^{-n\pi u} \sin(n\pi v)$$

$$\text{Along } C_2, u=0, \phi = \sin(\pi v) \quad \therefore B_1 = 1, B_n|_{n \neq 1} = 0$$

$$\therefore \phi(u, v) = e^{-\pi u} \sin(\pi v) = -e^{-\pi u} \sin(-\pi v)$$

$$\therefore \phi(x, y) = e^{-\pi(1-\frac{y}{r})} \sin(-\frac{\pi x}{x^2+y^2})$$

5.7 2011 Exam

(a) The function

$$w = \frac{1-z}{z}$$

maps a point $z = x + iy$ in the complex x - y -plane to a point

$$w = u(x, y) + iv(x, y)$$

in the complex u - v -plane. Find x and y in terms of u and v .

(b) Let C_1 be the circle $|z - \frac{1}{2}| = \frac{1}{2}$, C_2 be the circle $|z - \frac{1}{4}| = \frac{1}{4}$ and C_3 be the straight line $y = 0$ in the x - y -plane.

Sketch the domain D enclosed by C_1 , C_2 and C_3 in the upper half of the complex x - y -plane.

(c) The function $\phi(x, y)$ satisfies the Laplace equation

$$\phi_{xx} + \phi_{yy} = 0$$

in the region bounded by C_1 , C_2 and C_3 described in (b) and is subject to the boundary conditions $\phi = 0$ on C_1 and C_2 , and

$$\phi = \sin\left(\frac{\pi(1-x)}{x}\right)$$

on C_3 .

Sketch the image of the region in this map.

What equation does ϕ satisfy after the mapping in part (a)?

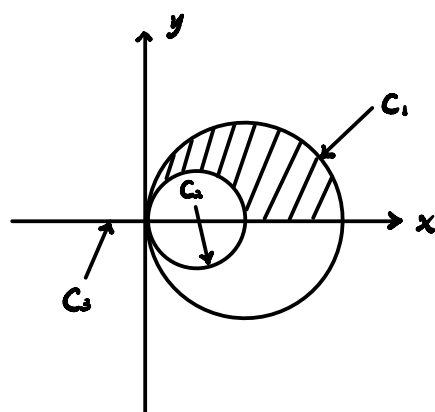
State the corresponding boundary conditions for ϕ in the u - v -plane.

(d) Solve the boundary-value problem in the u - v -plane by using the method of variable separation to obtain ϕ as a function of u and v , and hence find $\phi(x, y)$

$$(a) \quad w = \frac{1-z}{z} \quad \therefore zw = 1-z \quad \therefore z = \frac{1}{w+1}$$

$$\therefore x+iy = \frac{1}{(u+1)+iv} = \frac{(u+1)-iv}{(u+1)^2+v^2} = \frac{u+1}{(u+1)^2+v^2} + i \frac{-v}{(u+1)^2+v^2}$$

(b)



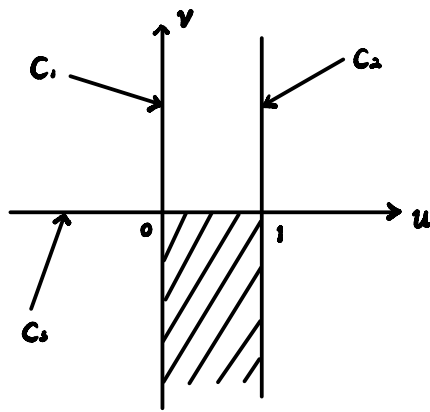
$$(c) \quad \text{for } C_1, \quad (x - \frac{1}{2})^2 + y^2 = (\frac{1}{2})^2 \quad \therefore x^2 + y^2 - x = 0$$

$$\therefore \frac{1-(u+1)}{(u+1)^2+v^2} = \frac{-u}{(u+1)^2+v^2} = 0 \quad \therefore C_1: u = 0 \text{ (imaginary axis)}$$

$$\text{for } C_2, \quad (x - \frac{1}{4})^2 + y^2 = (\frac{1}{4})^2 \quad \therefore x^2 + y^2 - \frac{1}{2}x = 0$$

$$\therefore \frac{1-\frac{1}{2}(u+1)}{(u+1)^2+v^2} = \frac{-\frac{1}{2}(u-1)}{(u+1)^2+v^2} = 0 \quad \therefore C_2: u = 1$$

$$\text{for } C_3, \quad y = 0 \quad \therefore \frac{-v}{(u+1)^2+v^2} = 0 \quad \therefore C_3: v = 0 \text{ (real axis)}$$



$$w = \frac{1-z}{z}$$

$$\therefore u + iv = \frac{(-x+1) - iy}{x+iy} = \frac{x - (x^2+y^2)}{x^2+y^2} + i \frac{-y}{x^2+y^2} = \left(\frac{x}{x^2+y^2} - 1\right) + i \frac{-y}{x^2+y^2}$$

$$\therefore \phi(0, v) = 0$$

$$\phi(1, v) = 0$$

$$\phi(u, 0) = \sin(\pi u)$$

(d) Let $\phi(u, v) = F(u) G(v)$

$$\therefore \nabla^2 \phi = 0$$

$$\therefore \frac{F''}{F} = -\frac{G''}{G} = -k^2$$

Let $F(u) = A \cos(ku) + B \sin(ku)$

$$G(v) = C e^{kv} + D e^{-kv}$$

$$\therefore \phi(0, v) = 0 \quad \therefore A = 0$$

$$\therefore \phi(1, v) = 0 \quad \therefore \sin(ku) = 0 \quad \therefore k = \pi n \text{ where } n = 0, 1, 2, \dots$$

$$\therefore \text{the region is bounded when } v \rightarrow \infty \quad \therefore C = 0$$

Let $D = 1$

$$\therefore \phi(u, v) = \sum_{n=0}^{\infty} B_n \sin(n\pi u) e^{-n\pi v}$$

By observation, $B_1 = 1$, $B_n|_{n \neq 1} = 0$

$$\therefore \phi(u, v) = \sin(\pi u) e^{-\pi v}$$

$$\therefore \phi(x, y) = \sin\left(\frac{\pi x}{x^2 + y^2} - \pi\right) e^{-\frac{\pi y}{x^2 + y^2}}$$

5.8 2020 Exam

- (a) The scalar variable ϕ satisfies Laplace's equation with respect to the Cartesian coordinates u and v ,

$$\frac{\partial^2 \phi}{\partial u^2} + \frac{\partial^2 \phi}{\partial v^2} = 0.$$

R and Θ are polar coordinates with $u = R \cos \Theta$, $v = R \sin \Theta$.

Prove that ϕ satisfies the polar form of Laplace's equation,

$$\frac{\partial^2 \phi}{\partial R^2} + \frac{1}{R} \frac{\partial \phi}{\partial R} + \frac{1}{R^2} \frac{\partial^2 \phi}{\partial \Theta^2} = 0.$$

- (b) Show that the radially symmetric solution $\phi = A \ln R + B$ satisfies

$$\frac{\partial^2 \phi}{\partial R^2} + \frac{1}{R} \frac{\partial \phi}{\partial R} + \frac{1}{R^2} \frac{\partial^2 \phi}{\partial \Theta^2} = 0$$

for real constants A and B , and find A and B given that when $R = \frac{1}{2}$, $\phi = 0$ and when $R = 1$, $\phi = 1$.

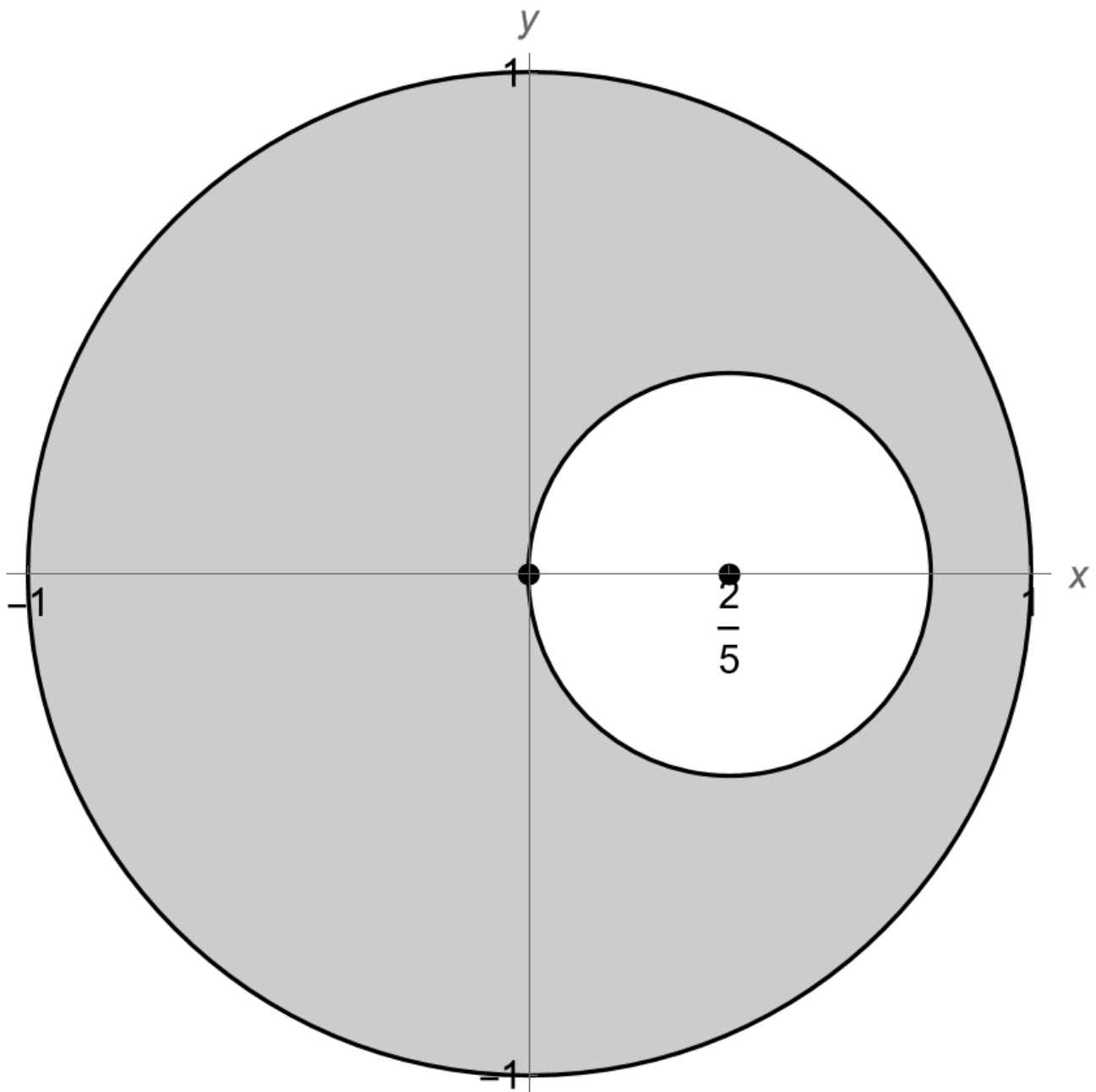
- (c) The map

$$w = \frac{2z - 1}{2 - z}, \quad z \neq 2$$

maps the complex variable $z = x + iy$ to the complex variable $w = u + iv$, where x , y , u and v are real. It may be assumed to be conformal everywhere in its domain.

By considering the value of $w, \bar{w}$ in the cases $z = e^{i\theta}$ and $\frac{2}{5}(1 + e^{i\theta})$, or otherwise, show that the images in this map of the circles $|z| = 1$ and $\left|z - \frac{2}{5}\right| = \frac{2}{5}$ are, respectively, the circles $|w| = 1$ and $|w| = \frac{1}{2}$.

- (d)



A cylinder of radius $\frac{2}{5}$ is held at an electrostatic potential of 0. It sits inside a larger cylinder of radius 1, held at an electrostatic potential of 1. A cross-section of the arrangement is shown in the xy -plane in the figure; the centre of the outer cylinder is at $(0, 0)$ and that of the inner cylinder at $\left(\frac{2}{5}, 0\right)$. The electrostatic potential ϕ in the region between the cylinders satisfies Laplace's equation, $\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$.

By transforming using the map described in part (c), express ϕ in terms of u and v .

(e) Hence, express ϕ in terms of x and y .

$$(a) \quad \frac{\partial \phi}{\partial R} = \frac{\partial \phi}{\partial u} \frac{\partial u}{\partial R} + \frac{\partial \phi}{\partial v} \frac{\partial v}{\partial R} = \frac{\partial \phi}{\partial u} \cos \theta + \frac{\partial \phi}{\partial v} \sin \theta$$

$$\frac{\partial \phi}{\partial \theta} = \frac{\partial \phi}{\partial u} \frac{\partial u}{\partial \theta} + \frac{\partial \phi}{\partial v} \frac{\partial v}{\partial \theta} = -\frac{\partial \phi}{\partial u} R \sin \theta + \frac{\partial \phi}{\partial v} R \cos \theta$$

$$\therefore \frac{\partial^2 \phi}{\partial R^2} = \frac{\partial}{\partial R} \left(\frac{\partial \phi}{\partial u} \cos \theta + \frac{\partial \phi}{\partial v} \sin \theta \right)$$

$$= \left(\frac{\partial}{\partial u} \frac{\partial u}{\partial R} + \frac{\partial}{\partial v} \frac{\partial v}{\partial R} \right) \left(\frac{\partial \phi}{\partial u} \cos \theta + \frac{\partial \phi}{\partial v} \sin \theta \right)$$

$$= \frac{\partial^2 \phi}{\partial u^2} \cos^2 \theta + 2 \frac{\partial^2 \phi}{\partial u \partial v} \cos \theta \sin \theta + \frac{\partial^2 \phi}{\partial v^2} \sin^2 \theta$$

$$\frac{\partial^2 \phi}{\partial \theta^2} = \frac{\partial}{\partial \theta} \left(-\frac{\partial \phi}{\partial u} R \sin \theta + \frac{\partial \phi}{\partial v} R \cos \theta \right)$$

$$= -\frac{\partial \phi}{\partial u} R \cos \theta - \frac{\partial \phi}{\partial v} R \sin \theta + \left(\frac{\partial}{\partial u} \frac{\partial u}{\partial \theta} + \frac{\partial}{\partial v} \frac{\partial v}{\partial \theta} \right) \left(-\frac{\partial \phi}{\partial u} R \sin \theta + \frac{\partial \phi}{\partial v} R \cos \theta \right)$$

$$= -R \frac{\partial \phi}{\partial R} + \frac{\partial^2 \phi}{\partial u^2} R^2 \sin^2 \theta + 2 \frac{\partial^2 \phi}{\partial u \partial v} R^2 \cos \theta \sin \theta + \frac{\partial^2 \phi}{\partial v^2} R^2 \cos^2 \theta$$

$$\text{Therefore, } \frac{\partial^2 \phi}{\partial R^2} + \frac{1}{R} \frac{\partial \phi}{\partial R} + \frac{1}{R^2} \frac{\partial^2 \phi}{\partial \theta^2} = \frac{\partial^2 \phi}{\partial u^2} + \frac{\partial^2 \phi}{\partial v^2} = \nabla^2 \phi = 0$$

$$(b) \frac{\partial \phi}{\partial R} = \frac{A}{R}$$

$$\frac{\partial^2 \phi}{\partial R^2} = -\frac{A}{R^2}$$

$$\frac{\partial \phi}{\partial \theta} = \frac{\partial^2 \phi}{\partial \theta^2} = 0$$

$$\therefore \frac{\partial^2 \phi}{\partial R^2} + \frac{1}{R} \frac{\partial \phi}{\partial R} + \frac{1}{R^2} \frac{\partial^2 \phi}{\partial \theta^2} = 0$$

$$\begin{cases} 0 = A \ln \frac{1}{2} + B \\ 1 = A \ln 1 + B \end{cases}$$

$$\therefore \begin{cases} A = \frac{1}{\ln 2} \\ B = 1 \end{cases}$$

$$(c) w = \frac{2z-1}{2-z} \text{ for } z \neq 2$$

$$\therefore 2w - wz = 2z - 1$$

$$\therefore z(w+2) = 2w+1$$

$$\therefore z = \frac{2w+1}{w+2}$$

$$\begin{aligned}
 \therefore x + iy &= \frac{(2u+1) + 2iv}{(u+2) + iv} = \frac{(2u+1)(u+2) + 2v^2 + i2v(u+2) - iv(2u+1)}{(u+2)^2 + v^2} \\
 &= \frac{2u^2 + 5u + 2 + 2v^2}{(u+2)^2 + v^2} + i \frac{3v}{(u+2)^2 + v^2} \\
 &= 2 - \frac{3(u+2)}{(u+2)^2 + v^2} + i \frac{3v}{(u+2)^2 + v^2}
 \end{aligned}$$

for $|z| = 1$, $x^2 + y^2 = 1$

$$\therefore \left[2 - \frac{3(u+2)}{(u+2)^2 + v^2} \right]^2 + \left[\frac{3v}{(u+2)^2 + v^2} \right]^2 = 1$$

$$\therefore 4 - \frac{12(u+2)}{(u+2)^2 + v^2} + \frac{9}{(u+2)^2 + v^2} = 1$$

$$\therefore \frac{12u+15}{(u+2)^2 + v^2} = 3 \quad \therefore \frac{4u+5}{(u+2)^2 + v^2} = 1$$

$$\therefore (u+2)^2 + v^2 - 4u - 5 = 0$$

$$\therefore u^2 + v^2 = 1$$

$$\therefore |w| = 1$$

for $|z - \frac{2}{5}| = \frac{2}{5}$, $(x - \frac{2}{5})^2 + y^2 = (\frac{2}{5})^2$

$$\therefore \left[\frac{8}{5} - \frac{3(u+2)}{(u+2)^2 + v^2} \right]^2 + \left[\frac{3v}{(u+2)^2 + v^2} \right]^2 = \frac{4}{25}$$

$$\therefore \frac{64}{25} - \frac{\frac{48}{5}(u+2)}{(u+2)^2 + v^2} + \frac{9}{(u+2)^2 + v^2} = \frac{4}{25}$$

$$\therefore \frac{\frac{48}{5}u + \frac{51}{5}}{(u+2)^2 + v^2} = \frac{12}{5} \quad \therefore \frac{16u+17}{(u+2)^2 + v^2} = 4$$

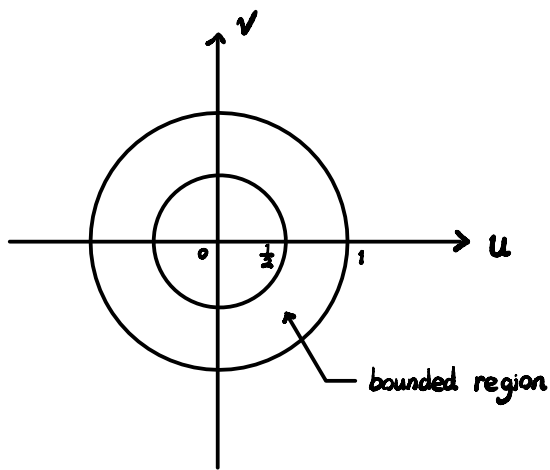
$$\therefore 4(u+2)^2 + 4v^2 - 16u - 17 = 0$$

$$\therefore 4u^2 + 4v^2 = 1$$

$$\therefore u^2 + v^2 = \frac{1}{4}$$

$$\therefore |w| = \frac{1}{2}$$

(d) $|w| = 1$ and $|w| = \frac{1}{2}$



$$\phi(|w|=1) = 1$$

$$\phi(|w|=1/2) = 0$$

$$\therefore \text{Let } \phi(u, v) = \phi(R, \theta)$$

$$\text{By inspection, } \phi(R, \theta) = \phi(R)$$

Therefore, according to (b),

$$\phi(R) = \frac{\ln R}{\ln 2} + 1 = \frac{1}{2\ln 2} \ln(u^2 + v^2) + 1$$

$$(e) \ w = \frac{2z-1}{2-z}$$

$$\therefore u + iv = \frac{(2x-1) + 2iy}{(-x+2) - iy} = \frac{-2x^2 + 5x - 2 - 2y^2 + 3iy}{(x-2)^2 + y^2} = -2 + \frac{3(x-2)}{(x-2)^2 + y^2} + i \frac{3y}{(x-2)^2 + y^2}$$

$$\therefore u^2 + v^2 = 4 - \frac{12(x-2)}{(x-2)^2 + y^2} + \frac{9}{(x-2)^2 + y^2} = 4 - \frac{12x-15}{(x-2)^2 + y^2} = \frac{(2x-1)^2 + 4y^2}{(x-2)^2 + y^2}$$

$$\therefore \phi(x, y) = \frac{1}{2\ln 2} \ln \left[4 - \frac{12x-15}{(x-2)^2 + y^2} \right] + 1 = \frac{1}{2\ln 2} \ln \left[\frac{(2x-1)^2 + 4y^2}{(x-2)^2 + y^2} \right] + 1$$